

Graph based Makeup Transfer

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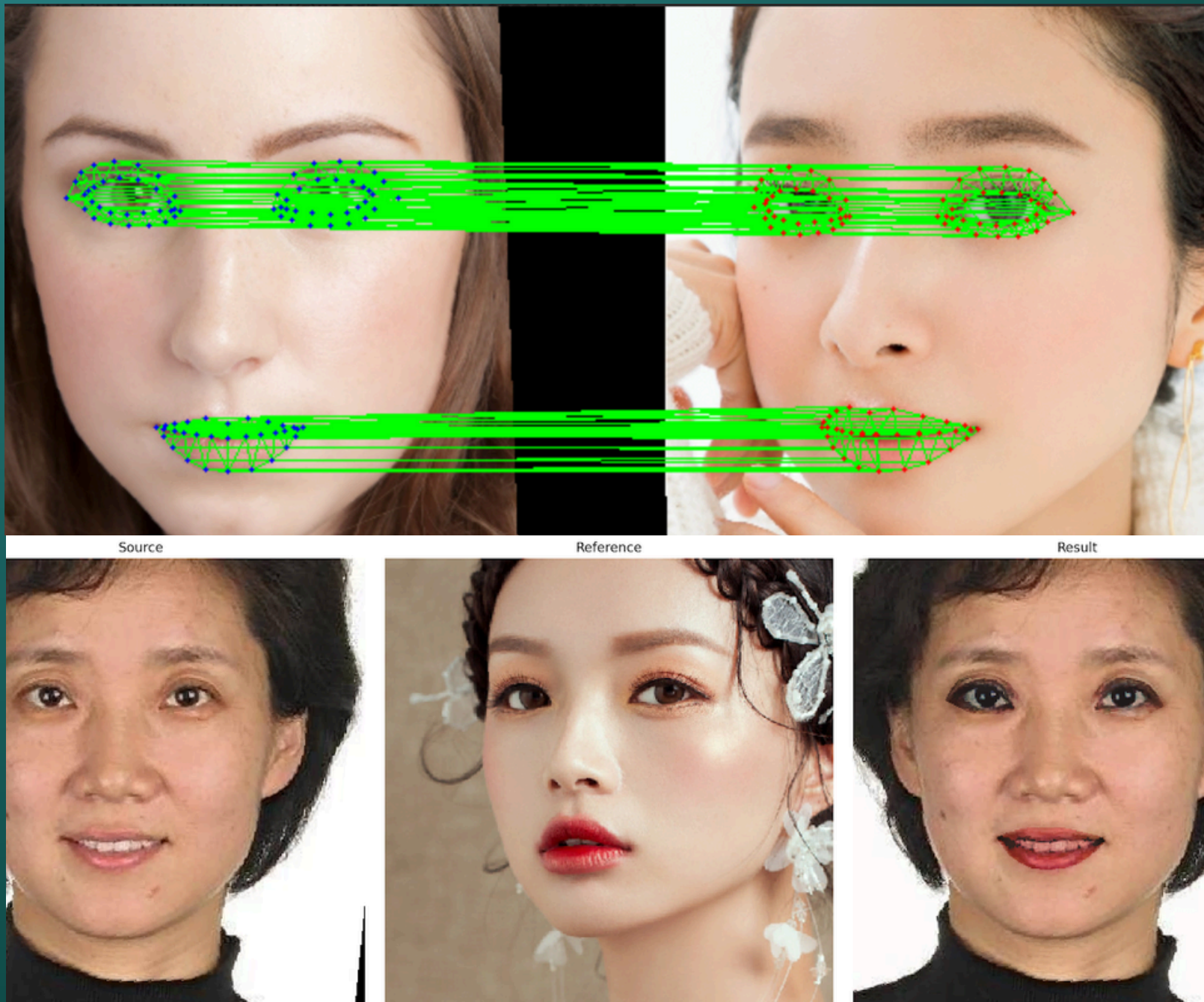
<https://github.com/sinhajiya/graphglow.git>

INTRODUCTION

Makeup transfer applies cosmetic styles from a reference face to a target face keeping the target's face identity and natural skin tone are preserved.

We represent the face as a graph built from meaningful landmark regions such as the lips and eyes.

A GCN enables structured, region-aware propagation of makeup features across this graph, producing smoother and more consistent transfers.



DATASET



Figure 3: Samples from MT dataset. The non-makeup and makeup images are shown in upper row and lower row.

Makeup transfer (MT):

- 1115 non-makeup images
- 2719 makeup images

FACE AS GRAPH

Main region of focus for makeup:

- Lips (lipstick)
- Eyes (eye liner, eye shadow)

Isolate and model only these regions

Nodes:

Individual facial landmark points within each region of interest (lips, left eye, right eye).

Edges:

- intra-face edges: kNN
- inter-face edges: corresponding landmarks connected



WHY FACE AS GRAPH

Pose invariant

While pixel coordinates change drastically with pose, landmark does not.

Controlled information flow

Edges let makeup flow only along meaningful facial geometry.
Reduces background noise.

NODE FEATURES

LAB std + LAB mean + RGB mean

What exactly should be 'transferred' in makeup?

Color information, luminance, hue, undertones, and tiny local contrasts.

Why not just RGB?

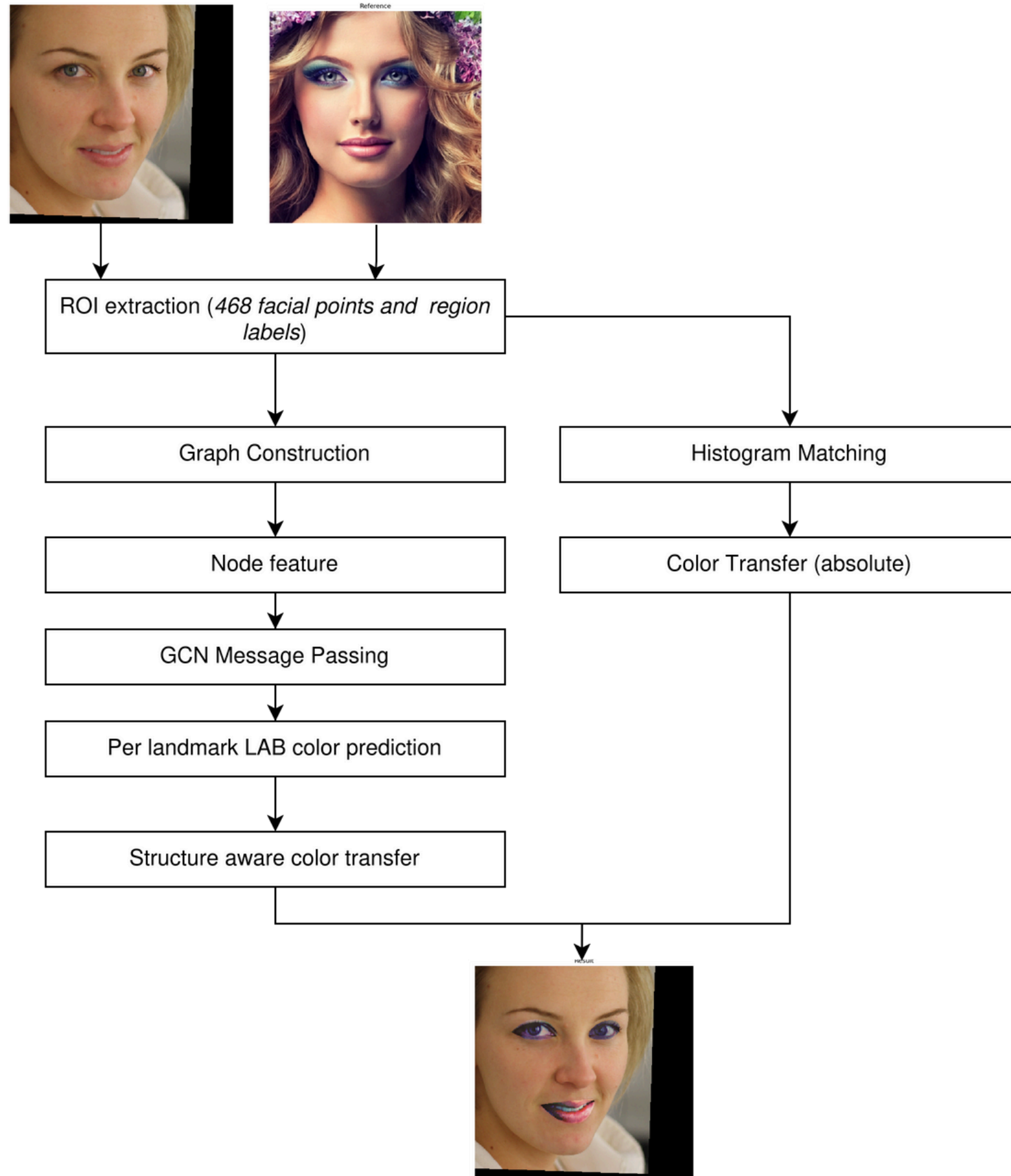
RGB averages miss perceptual color differences and texture.

Why Lab mean + LAB Std Dev?

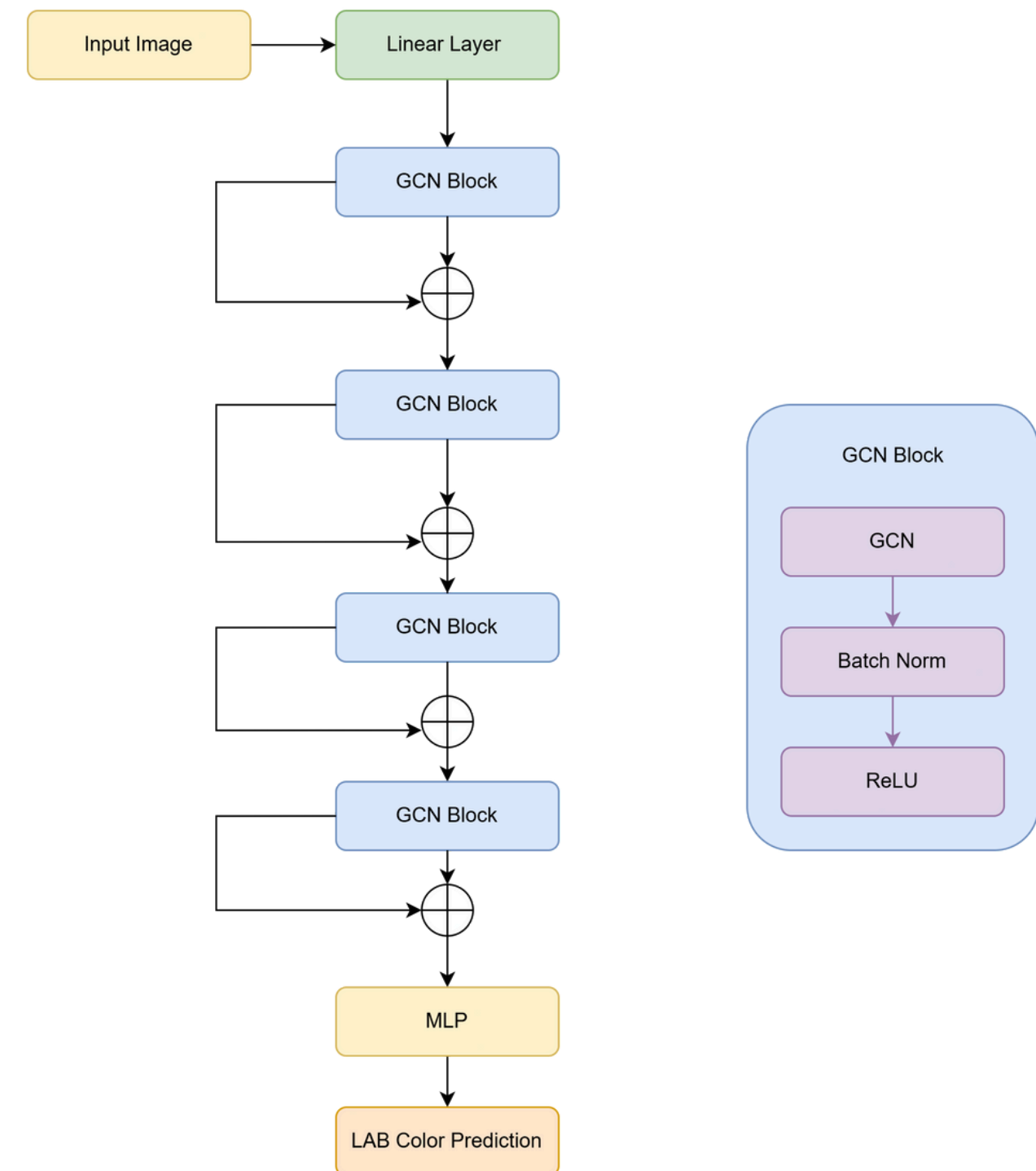
Lab captures perceptual lightness and hue.

Std dev captures texture, glossiness, and contrast

Makeup transfer Network



Graph-Based Color Regression Network



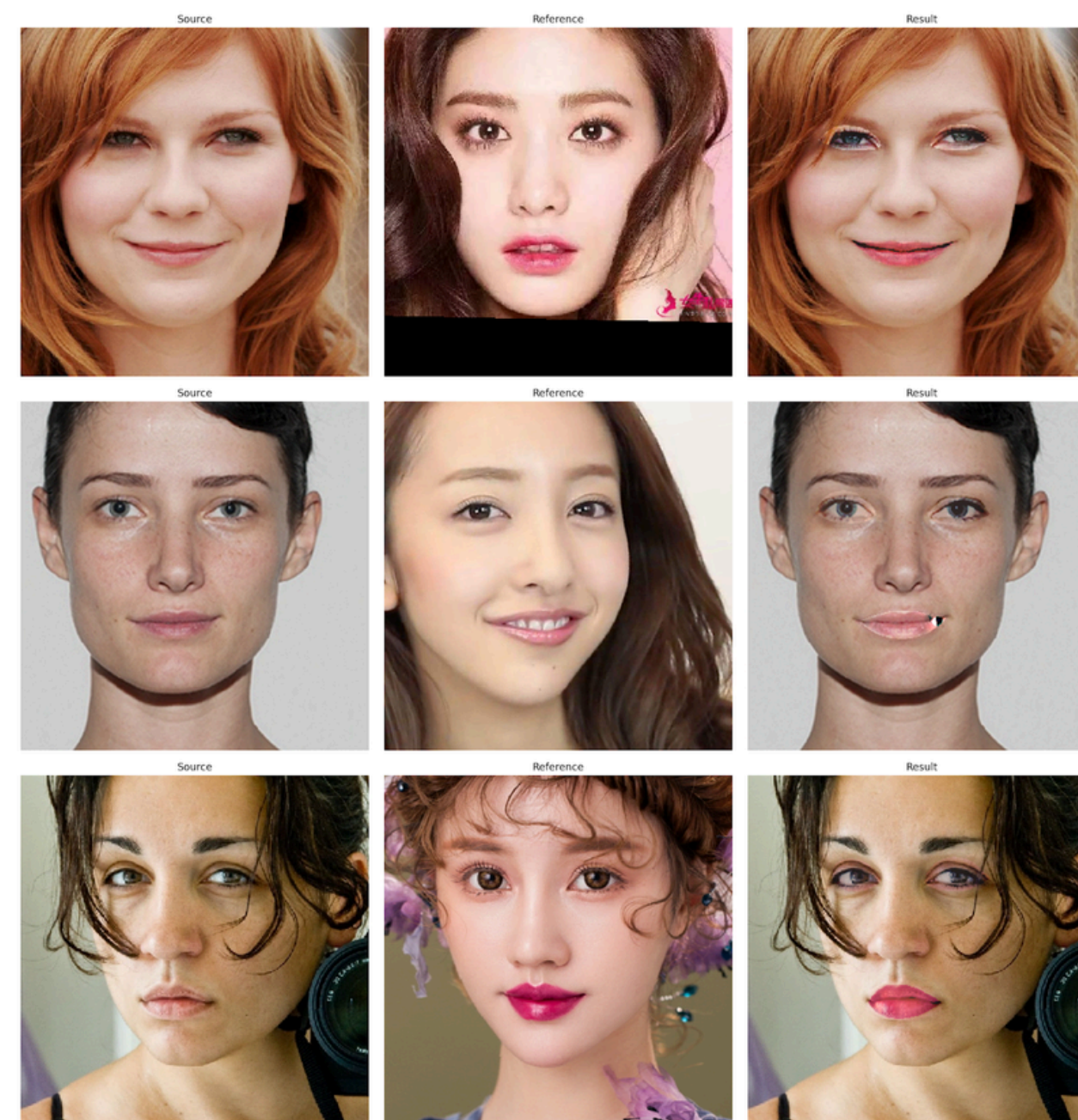
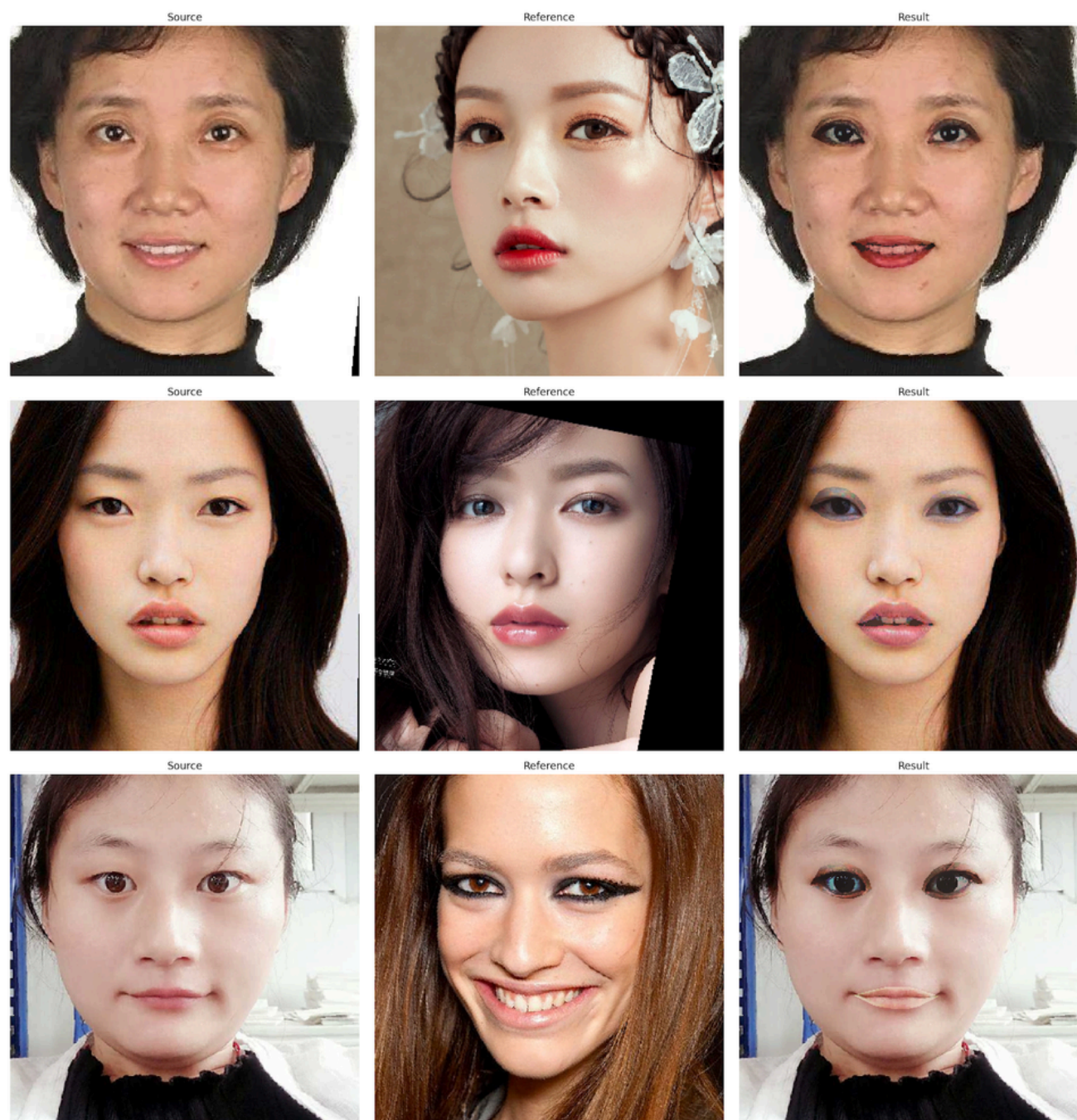
Propagate color information across the graph, then predicts the final LAB color for each landmark.

It learns how makeup color should flow across facial geometry and between corresponding landmarks of two faces.

EVALUATION

Through user reviews

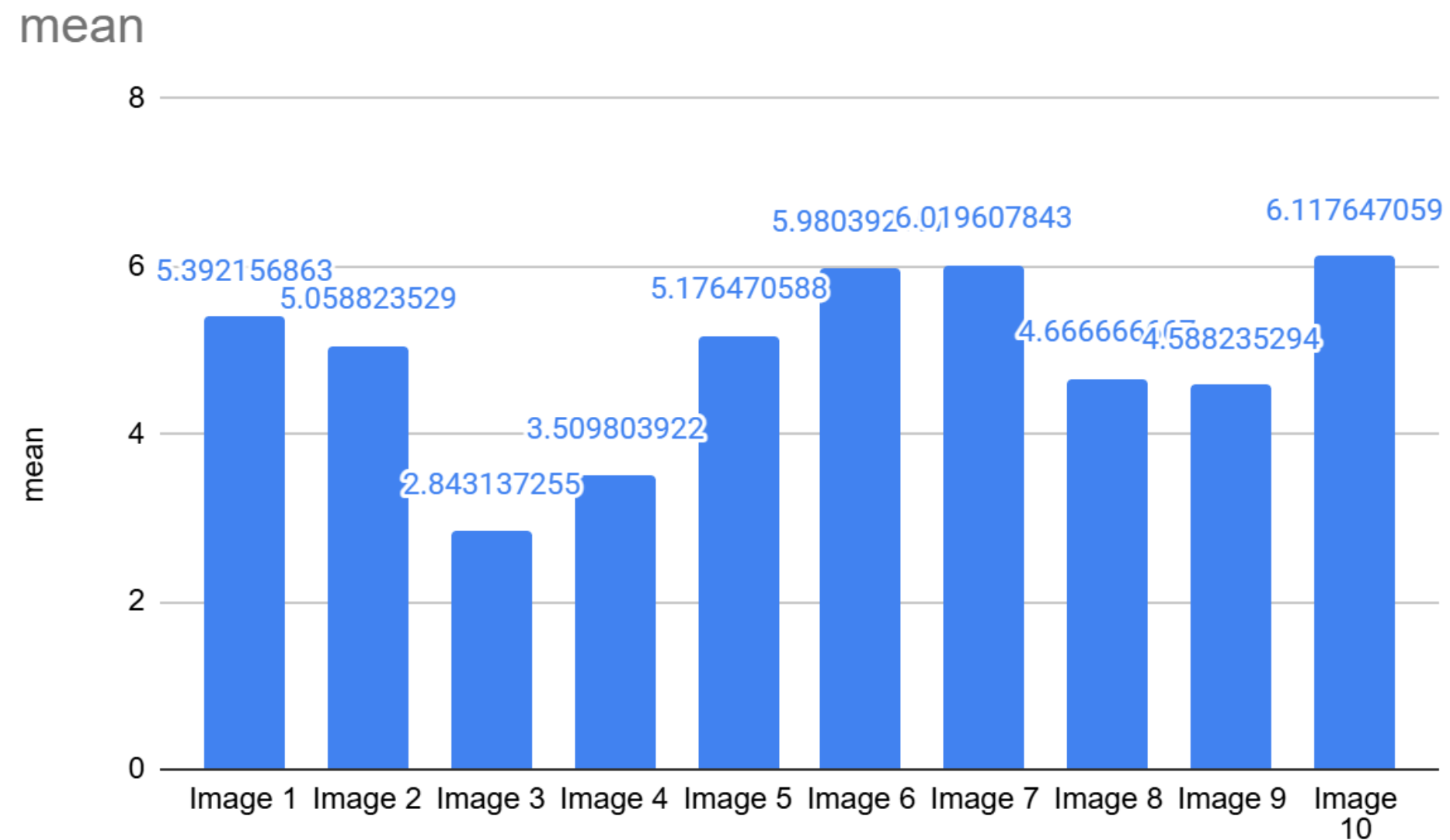
Randomly sampled 10 predicted images and asked 54 users to rate on the scale of 1-10.



EVALUATION

Through user reviews

Mean of scores of individual images



IMAGES	MEAN OF SCORES
IMAGE1	5.392156863
IMAGE2	5.058823529
IMAGE3	2.843137255
IMAGE4	3.509803922
IMAGE5	5.176470588
IMAGE6	5.980392157
IMAGE7	6.019607843
IMAGE8	4.666666667
IMAGE9	4.588235294
IMAGE10	6.117647059

Mean Score = 4.9/10

TOTAL VIEWERS=54
54 GOOGLE RESPONSES RECIEVED

IMAGES WHICH GOT HIGH RATINGS FROM THE VIEWERS

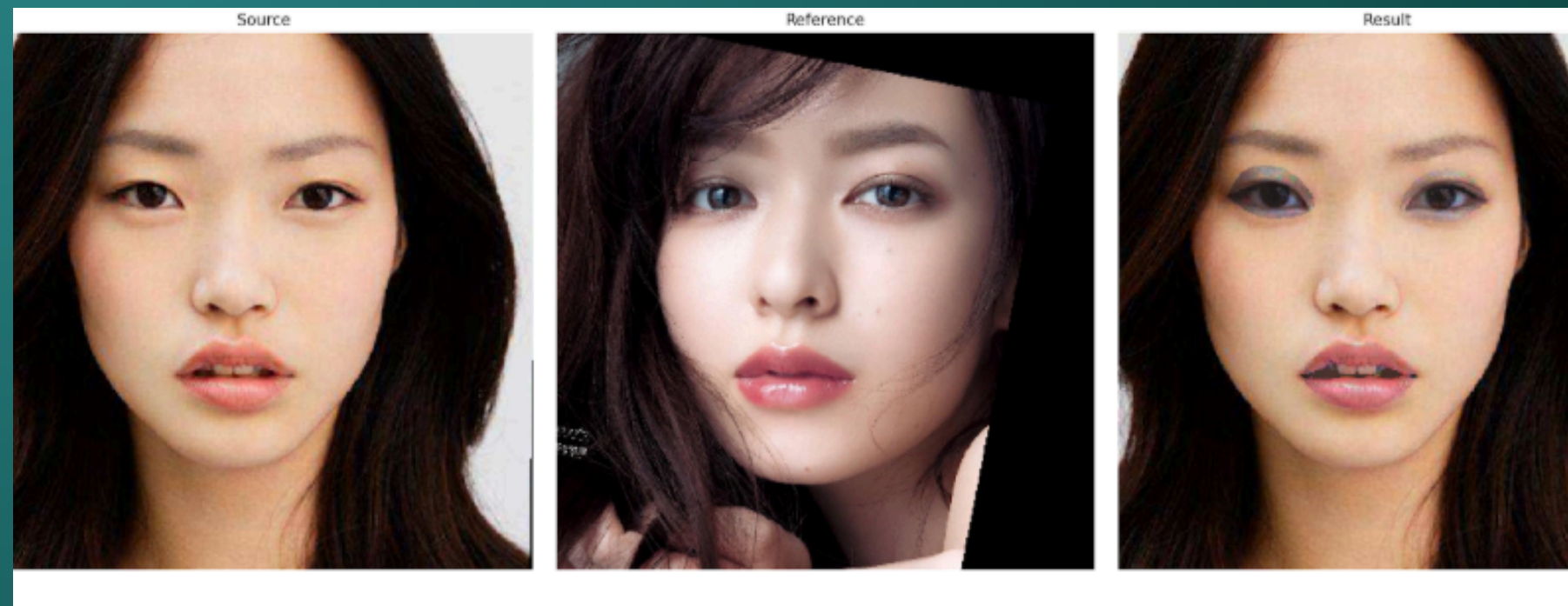


IMAGE 10



IMAGE 6

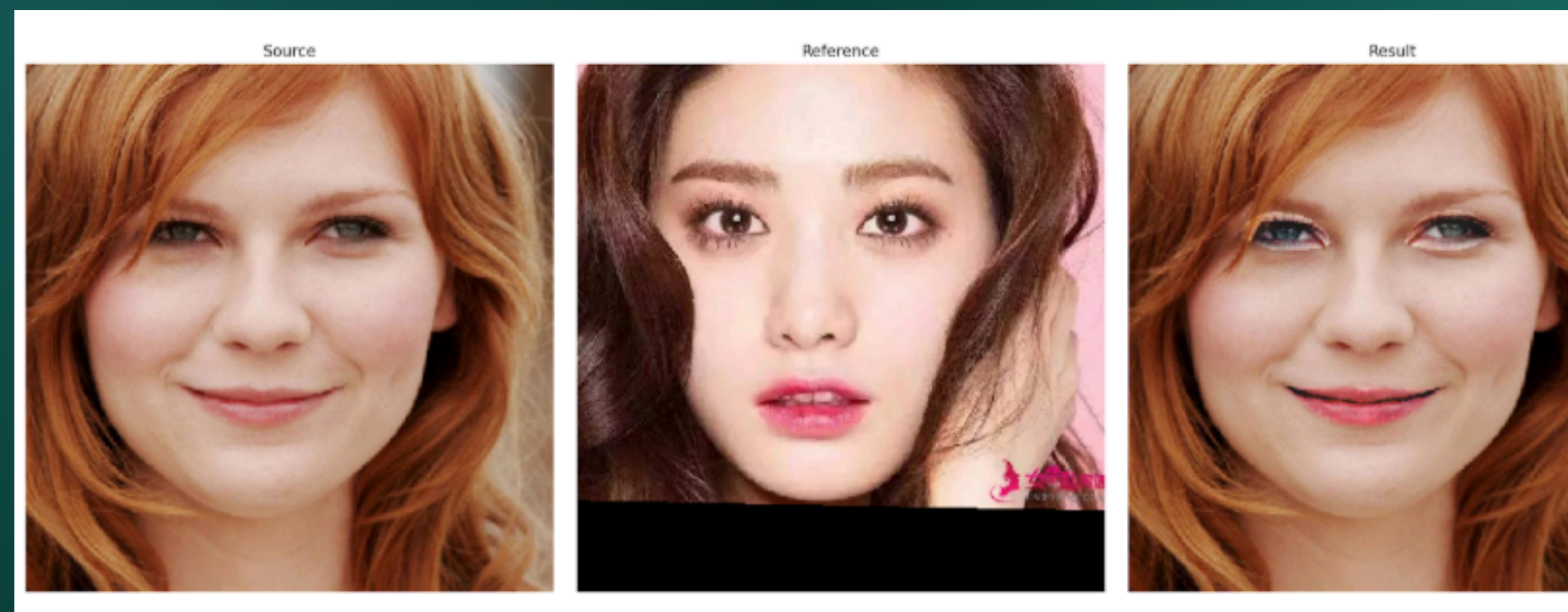
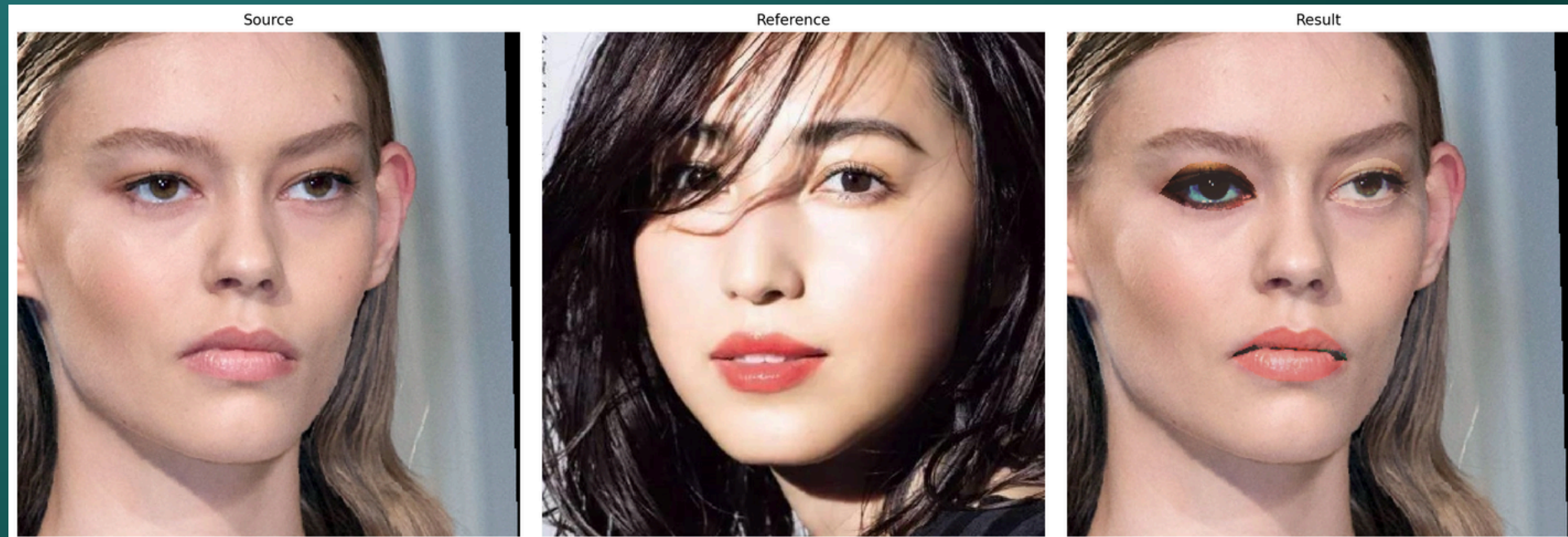


IMAGE 7

LIMITATIONS



Occlusions confuse the model: eg hair over the face gets colored as if it were the makeup

Cheek makeup (blush/contour) isn't handled well due to pre-defined set of landmarks

Thank You!